

Kyle Main

Dallas/Ft. Worth Area | 469-545-3791 | main.kyle87@gmail.com

Machine Learning Engineer — Applied AI & Security Data Science

ML engineer and security data scientist with 12 years of experience (since 2015) training and evaluating ML models — clustering, time-series anomaly detection, and GenAI/LLM-powered automation — against security data at scale. Direct, hands-on experience creating and managing Cribl pipelines. Python/PyTorch/scikit-learn foundation, M.S. Physics (numerical modeling and data analysis).

CORE SKILLS

Cribl & Data Pipelines: Hands-on experience creating and managing Cribl pipelines for security/observability data routing; broader pipeline engineering across 220+ log sources (Logstash, Elasticsearch Beats, Apache Beam/GCP Dataflow)

GenAI / LLM Applications: Prompt engineering for security use cases (false-positive triage, detection-content generation); GenAI-driven orchestration of SIEM APIs across 9 platforms; built reusable GenAI-powered "skills" automating detection-rule conversion between SIEM syntaxes

ML & Data Science: Python, PyTorch, scikit-learn, Pandas, NumPy, SciPy; trained/evaluated unsupervised clustering models (network device behavior); time-series anomaly-detection models (auth/process-chain behaviors); large-scale EDA via PySpark, GCP Dataproc, SparkSQL

Platforms & Engineering: Elasticsearch (queries, transforms, native detection rules, API), Kibana, Splunk; GitLab CI/CD pipeline with automated testing and staged rollout; Docker; AWS, GCP, Azure

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Treasury SOC (TSSOC), Threat & Research team (current): build and maintain Splunk-based detection/analytics content the SOC runs incident response against.
- DOE/NNSA Security Data Integration (completed): built a new Elasticsearch-based security data platform ingesting CrowdStrike, Suricata, and Zeek; built the UEBA detection layer on ES transforms, plus dashboards and data-quality monitoring.
- CISA CDM at DOE (completed): data ingestion and quality engineering across a combined Elasticsearch and Splunk environment.

Senior Threat Detection Engineer & Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of the detection engineering/data-science team; built signature, behavioral, statistical, time-series, and ML-based detection models against cloud-scale customer data.

Threat Detection Engineer & Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Trained and evaluated unsupervised ML models for network device clustering as part of a next-gen SIEM's rules engine spanning 2,300+ detection rules across the MITRE ATT&CK matrix.
- Built time-series anomaly-detection models for entity behavior — authentication attempts by country/volume, parent/child process chains, Outlook process-chain analysis.
- Ran exploratory data analysis at scale on GCP Dataproc (PySpark/SparkSQL, Zeppelin notebooks); built GenAI-powered tooling automating cross-SIEM detection-rule conversion and false-positive triage — production LLM tooling, not a research exercise.
- Data engineering: 220+ ingested log sources, 50+ Logstash filters, Common Information Model (CIM) design, GCP Dataflow/Apache Beam for cold-storage retrieval.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Built DNS-based detection and mitigation models for network malware infections; analyzed large-scale security log data to surface anomalous behavior.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013, Numerical Data Analysis & Modeling) | B.S. Physics — Ball State University (2011; Minors: Mathematics, Astrophysics)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox